

Lucas Lu

B.Eng. Candidate in Industrial Engineering | Class of 2029

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RESEARCH INTERESTS & MENTORSHIP

Interests: Operations Research & Optimization; Emergency Response & Logistics; Machine Learning & Data Science; Mathematical Modeling.

Academic Mentor: Professor Yu Wang, Department of Industrial Engineering, SUSTech.

EDUCATION

Southern University of Science and Technology (SUSTech)

Aug. 2025 - Present

B.Eng. in Industrial Engineering, Zhicheng College | Shenzhen, China

GPA: 3.91 / 4.0 | Ranking: 1 / 11.

Selected Coursework: Mathematical Analysis I-II (93, 96), Ordinary Differential Equations B (98), Foundation of Probability Theory (94), Linear Algebra (89), Introduction to C Programming (96), College Physics II (90).

SELECTED RESEARCH & PROJECTS

EMS Station Location Optimization

2026 - Present

Project Lead | 120 EMS Station Location Optimization

Leading a 6-member team to optimize emergency medical service station placement across Shenzhen. Partitioned the city into 5,279 500m x 500m grids and developed an XGBoost ETA model using 620K+ dispatch records and 360K Tencent Maps samples (MAE = 1.64 min, R2 = 0.863); achieved 90% coverage within 10 minutes.

Finalist, 8th China University Mechanical Engineering Innovation & Creativity Competition (2026). Tools: Python, OSMnx, XGBoost.

AED Deployment at Gulongzhong

2026 - Present

Project Participant | AED Deployment at Gulongzhong

Comparing static AED placement with a dynamic human-vehicle-drone coordinated scheme using road-network analysis; exploring sustainable operation through insurance partnership models.

Meal Optimization WeChat Mini Program

2026

Project Developer | Personalized Meal Recommendation

Developed a WeChat mini program for personalized meal recommendations from user profiles and dietary preferences, using Taro + React + TypeScript with Tencent Cloud Functions and an optimization solver. Supported meat/vegetable/staple structure configuration, taste matching, nutrition constraints, and set-meal recommendations.

Implemented upper/lower bounds for calories, protein, carbohydrates, and fats; hard-excluded disliked ingredients, dish-level replacement, infeasibility fallbacks, and carbohydrate-coverage alerts with adjustment suggestions. Improved stability and UX through legacy solver compatibility, exception fallback, and frontend feedback.

SELECTED HONORS & AWARDS

First Prize, Guangdong-Hong Kong-Macao Greater Bay Area Industrial Engineering Innovation Competition - 2026

APRU ULP Outstanding Student Ambassador (one of ten university-wide), APRU / SUSTech - 2026

Second Prize, 17th National College Student Mathematics Competition (Non-Math A) - 2025

Outstanding Individual, SUSTech Winter Social Practice - 2026

Outstanding Camper, 4th Xiancheng Program, Zhicheng College - 2025

Social Impact Award, Hundreds, Thousands, Myriads Project, Chengguang Volunteer Team - 2025

Outstanding Volunteer Service Organization of 2025, Chengguang Volunteer Team - 2025

LEADERSHIP & SERVICE

Core Student Leader | SUSTech IE Hunt

Oct. 2025 - Present

Optimization challenge planning and student team coordination

Co-led Seasons 1-2 of a campus-wide optimization challenge, integrating the Traveling Salesman Problem and Linear Programming into game mechanics; coordinated a 5-member team across logistics, feasibility analysis, and promotion.

Student Representative | Zhicheng College 10th Anniversary Ceremony

Oct. 2025

Selected as the sole freshman representative to deliver a formal keynote speech

Volunteer | APEC Shenzhen 2026

2026

Volunteer Service

Group Leader | Orange Light Volunteer Service Team

2025 - Present

Community research and accessibility advocacy

Completed 160+ hours of volunteer service. Led community research across Shenzhen, Foshan, and Guangzhou; partnered with the Shenzhen Association for the Blind on accessibility supervision and hosted Oxford University delegations at SUSTech.

SKILLS

Programming & Technical: Python, MATLAB, R, C, LaTeX, AnyLogic, XGBoost, scikit-learn, NumPy, Pandas, Matplotlib, OSMnx.

Languages: Mandarin (Native), English (Proficient), Korean (Daily basics).